DEXTROMETHORPHAN

Serotonin syndrome risk

Dry cough only

CHEMICAL NAME (+)-3-methoxy-17-methyl-9α,13α,14α-morphinan (hydrobromide)
CLASS Non-opioid antitussive (cough suppressant)
TRADE NAMES Robitussin DM · Benylin Dry Cough · Delsym · Tusscalm

THERAPEUTIC USES / INDICATIONS

- Symptomatic relief of dry, non-productive cough.
- Combined with bupropion/quinidine for specialised neurological indications.

PHARMACODYNAMICS

- Acts centrally on **cough centre** in medulla (sigma receptor agonist, NMDA antagonist).
- No analgesic or respiratory-depressant effect at normal doses.

ADMINISTRATION

- **PO** syrup or tablets; use the measuring device provided.
- Not for productive (wet) cough — sputum retention.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dizziness, drowsiness.
- Nausea, GI upset.

ADVERSE EFFECTS (SERIOUS)

- Serotonin syndrome with SSRIs/MAOIs.
- Abuse at high doses (dissociation, hallucinations).
- Respiratory depression in overdose.

DOSAGES

- **Adult:** 10–20 mg every 4 h or 30 mg every 6–8 h; max 120 mg/day.
- **Children 6–12 yr:** 5–10 mg every 4 h (max 60 mg/day).
- Not recommended <6 yr (OTC).

CONTRAINDICATIONS & CAUTIONS

- MAOIs within 14 days.
- Children <6 yr (OTC).
- **Caution:** asthma, productive cough, liver disease.

KEY INTERACTIONS

- **MAOIs, SSRIs (fluoxetine, paroxetine), linezolid:** serotonin syndrome.
- **CYP2D6 inhibitors:** ↑ levels.
- **Alcohol, sedatives:** ↑ CNS depression.

PHARMACOKINETICS

- **Onset:** 15–30 min. **Duration:** 3–6 h.
- Metabolised by CYP2D6 to dextrorphan.
- Excreted in urine. **t½:** 2–4 h (longer in poor metabolisers).

NURSING MANAGEMENT

- Assess cough type; avoid in productive cough.
- Check antidepressant use.
- Evaluate cough duration — >3 weeks needs investigation (e.g. TB).

PATIENT EDUCATION

- Use for dry cough only.
- Do not combine with other cough/cold products or antidepressants without advice.
- Seek review if cough lasts >3 weeks or blood in sputum.

KEY POINTS

- Central antitussive for dry cough.
- Max 120 mg/day.
- Serotonin syndrome with SSRIs/MAOIs.
- Abuse potential.